PVC Suspension Reactor Ver 1 · R-101 (one of four 140 m³ batch suspension-PVC reactors)

# Procurement, completion and commissioning plan

## PVC-V1-PM-CSU-001

Procurement, completion and commissioning plan for R-101 generated from the PVC Suspension Reactor Ver 1 OTS: 11 packages, 9 completion systems, 12 commissioning steps mapped to the OTS procedures and SOPs, the performance test from the reference batch (5.27 h, X 0.848, 43.2 t PVC), 9 of 10 cross-checks passing and 4 engineering queries.

| | |
|---|---|
| **Unit** | PVC Suspension Reactor Ver 1 · R-101 (one of four 140 m³ batch suspension-PVC reactors) |
| **Prepared for** | Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University |
| **Document no.** | PVC-V1-PM-CSU-001 |
| **Title** | Procurement, completion and commissioning plan |
| **Status** | Rev 0 · Issued for teaching |
| **Originator** | Apinan Soottitantawat |

| Rev | Date | Description | Originator | Checked | Approved |
|---|---|---|---|---|---|
| 0 | 2026-10-27 | Issued for teaching | Apinan Soottitantawat | | |

The numbers in this document come from the PVC Suspension Reactor Ver 1 simulator model (R-101) and the documents of the same OTS, or are labelled "assumed" or "typical". It was generated for teaching: it is not a design deliverable and must not be used for construction or purchase. Checked and Approved are left blank on purpose.

# 1 Scope and basis

This plan covers reactor R-101 of PVC Suspension Reactor Ver 1 from the purchase orders to handover: 11 procurement packages with their inspection and test plans, FAT and SAT with the OTS model as the plant emulator, 9 completion systems with their checklists, 12 commissioning steps each mapped to the OTS procedure or SOP that rehearses it, and the performance test. Equipment comes from PFD-100, instruments from PVC-V1-IN-INL-001, trips and effects from PVC-V1-IN-CE-001 and PVC-V1-IN-SRS-001, procedures from the OTS and the SOP register; the guarantees from the model's reference batch.

• Lead times, durations, ITP content, acceptance tolerances and the three-batch performance test are typical (assumed).

• Standards quoted as typical practice: ASME VIII Div. 1 (pressure test), API 610 (pumps), IEC 61511 (SIS validation and proof test), ISA-18.2 (alarms), EN 10204 (material certificates).

• No supplier is named: packages are scopes, not orders.

# 2 Phases and gates

| Phase | Main activities | Gate |
|---|---|---|
| **Procurement** | Purchase orders, supplier documents, ITPs, shop inspection and FAT | PO → equipment shipped |
| **Construction** | Erection, piping, cabling, insulation | Mechanical completion (MC) certificate per system |
| **Pre-commissioning** | Flushing, leak and vacuum tests, loop checks, SIS proof tests, motor runs | Ready for commissioning |
| **Commissioning** | Water batch, ESD test, inerting, VCM introduction, first batch | Ready for start-up / first batch |
| **Performance test** | Consecutive batches against the guarantees | Provisional acceptance |
| **Handover** | Punch B closed, documents as built, crew trained | Operations take over |

# 3 Procurement packages

| Package | Scope | Items | Lead time | Shop test / FAT |
|---|---|---|---|---|
| **PK-01** Reactor R-101 with agitator M-101 | Agitator M-101 (motor, gearbox, 2 impellers, double mechanical seal); Jacket and four water-cooled baffles; Wash nozzles, nozzles for TE-100/TE-101, PT-100/PT-101, LT-100 | R-101 PVC Reactor 140 m3 | 52 wk | Agitator mechanical run test with the vendor motor |
| **PK-02** Reflux condenser E-101 | Vertical, VCM in the tubes, CW on the shell | E-101 Reflux Condenser | 30 wk | — |
| **PK-03** Pumps | Jacket circulation pumps P-101A/B; VCM charge pump P-103 (sealless or double seal); Initiator dosing pump P-104; Slurry pump P-102 | P-101 A/B Jacket Circulation Pumps; P-103 VCM Charge Pump; P-104 Initiator Dosing Pump; P-102 Slurry Pump | 26 wk | Witnessed performance test of P-101A/B |
| **PK-04** Vacuum pump and VCM recovery compressor | VP-101 evacuates R-101 to 0.03 bar a; C-101 returns VCM to the gasholder D-106 at 1.25 bar a | VP-101 Vacuum Pump; C-101 VCM Recovery Compressor | 36 wk | Mechanical run and vacuum test |
| **PK-05** Pressure vessels and tanks | VCM storage, PVA pot, initiator pot (chilled), short-stop pot ($N_2$), emergency vent tank, gasholder, blowdown tank | TK-100 VCM Storage; TK-107 PVA Pot; D-104 Initiator Pot (chilled); D-105 Short-stop Pot (N2); D-103 Emergency Vent Tank; D-106 VCM Gasholder 1.25 bar a; D-102 Blowdown / Slurry Tank | 30 wk | — |
| **PK-06** Instruments | Transmitters, elements and thermowells, flowmeters, load cells, regulators | 18 lines: FT-102, FT-101, FT-103, JT-100, PT-100, LT-100, TT-100, TT-111, TT-110, FT-120, TT-121, FT-130, TT-131, TT-132, TT-120, WT-100, PCV-141, PCV-112 | 16 wk | — |
| **PK-07** Control and on-off valves | Control valves with HART positioners, on-off valves with solenoids and limit switches | 15 lines: FV-101, FV-102, FV-103, TV-110A, TV-110B, FV-120, FV-130, HV-135, PV-105, XV-101, XV-102, XV-103, XV-105, XV-106, XV-109 | 22 wk | Fail-action witness on FV-130 / TV-110B (fail open) and XV-104 (energise to open) |
| **PK-08** Safety instrumented system | SIS logic solver (independent of the DCS), SIS I/O, ESD push-button HS-001 | 8 lines: TT-101, FT-110, PT-101, XV-104, XV-108, XS-101, HS-001 | 30 wk | SIS FAT against PVC-V1-IN-CE-001 with the OTS model as the plant emulator |
| **PK-09** Distributed control system | Controllers, I/O, operator stations, alarm management | 72 BPCS I/O installed | 30 wk | DCS FAT with the OTS model as the plant emulator (tie-back) |

| Package | Scope | Items | Lead time | Shop test / FAT |
|---|---|---|---|---|
| **PK-10** Electrical: MCC, motors, cables | MCC for M-101, P-101A/B, P-103 and the package motors; power and instrument cables | 4 lines: M-101, P-101A, P-101B, P-103 | 28 wk | MCC FAT |
| **PK-11** Shared downstream units (interface only) | Shared by R-101 to R-104; tie-ins and interface checks only | T-201 Slurry Stripping Column; S-201 Dewatering Centrifuge; D-201 PVC Dryer | — | — |

## 3.1 Inspection and test plans

| Package | Inspection and test | Point |
|---|---|---|
| **PK-01** | Material certificates EN 10204 3.1 for the clad plate | W |
| | Weld procedures, NDT (RT/UT) of the shell and jacket welds | W |
| | Shop hydrostatic test of the shell (1.3 × design pressure, ASME VIII Div. 1 UG-99, typical) and of the jacket | H |
| | Agitator: mechanical run in the shop, seal leak test, vibration | W |
| | Internal finish (polished, anti-fouling) inspection | W |
| | Dimensional check of nozzles and baffles | W |
| **PK-02** | Tube-to-tubesheet weld and leak test | W |
| | Hydrostatic test of both sides | H |
| | Tube internal finish (polymer build-up) | W |
| **PK-03** | Performance test on water (head, flow, power, NPSH), API 610 (typical) | H |
| | Seal test | W |
| | Vibration and noise | W |
| **PK-04** | Mechanical run test | W |
| | Ultimate vacuum test (VP-101) | W |
| | Gas-tightness test with helium (VCM service) | W |
| **PK-05** | Material certificates, NDT, hydrostatic test | H |
| | Short-stop pot D-105: $N_2$ pressure test and level gauge | W |
| **PK-06** | Calibration certificates at the INL range | W |
| | Thermowell wake-frequency calculation | W |
| | Hazardous-area certificates | W |
| **PK-07** | Seat leakage test (FCI 70-2 class, typical) | W |
| | Stroke test and fail-action test (air off → FC / FO) | H |
| | Solenoid and limit-switch function | W |
| **PK-08** | SIL certificates of the logic solver and devices | H |
| | Application software against the C&E and the SRS | H |
| **PK-09** | Hardware FAT (I/O, redundancy, power) | H |
| | Software FAT: loops, sequence, displays, alarms | H |
| **PK-10** | Motor routine tests | W |
| | MCC FAT (interlocks, protection settings) | H |
| **PK-11** | Tie-in and interface check | W |

## 3.2 FAT and SAT with the OTS as the plant emulator

The DCS and SIS application software is tested against the OTS model (the same R-101 equations the trainees use) through a soft I/O link instead of loop-back wires, so every test runs against a reactor that responds. After SAT the emulator stays on as the training simulator.

| Test | Scope | Count | How the OTS is used |
|---|---|---|---|
| SIS FAT | Every C&E trip and permissive | 30 causes, 114 effect cells, 6 trip inputs | Force each cause on the emulator, check every effect and the first-out against PVC-V1-IN-CE-001 |
| DCS FAT: loops | Controllers, cascades, split range | 11 controllers | Step tests on the emulated reactor |
| DCS FAT: sequence | Batch sequence and permissives | 7 commands | OTS expert procedures as test scripts |
| DCS FAT: alarms | Limits, priorities, suppression | 11 alarm tags | Drive each variable across its limit |
| Procedures | Commissioning rehearsal | 6 procedures | Run against the real configuration before SAT |
| SAT | Field devices connected | section 4 | Repeat the FAT scripts with the real devices |

# 4 Completion systems and checklists

## SYS-01 Reactor R-101, agitator and baffles

**Equipment:** R-101, M-101. **Instrument list:** JT-100, PT-100, LT-100, TT-100, XI-100, WT-100, QI-100, PD-100, KX-100, LIC-100, TIC-100

| Stage | Checks |
|---|---|
| Mechanical completion | Equipment and piping: Installed to the P&ID and the GA: R-101, M-101; Equipment and piping: Supports, insulation, nameplates, punch A closed; Instruments and valves: Installed and tagged: JT-100, PT-100, LT-100, TT-100, WT-100 |
| Pre-commissioning | Equipment and piping: Flushed / cleaned, leak tested; Equipment and piping: Rotating equipment aligned and solo-run; Instruments and valves: Loop checks 5 instruments, 0 valves (range, alarms, trips, fail action) |
| Commissioning | Equipment and piping: In service on water (and VCM where it applies); Instruments and valves: Loops in AUTO / CAS at the design set points, tuned |
| Handover | Equipment and piping: Punch B closed, records signed; Instruments and valves: As-built loop records |

## SYS-02 Jacket tempered-water loop

**Equipment:** P-101A, P-101B. **Instrument list:** TT-111, TT-110, FT-110, TIC-110, TV-110A, TV-110B, TY-110, PCV-112

| Stage | Checks |
|---|---|
| Mechanical completion | Equipment and piping: Installed to the P&ID and the GA: P-101A, P-101B; Equipment and piping: Supports, insulation, nameplates, punch A closed; Instruments and valves: Installed and tagged: TT-111, TT-110, FT-110, TV-110A, TV-110B |
| Pre-commissioning | Equipment and piping: Flushed / cleaned, leak tested; Equipment and piping: Rotating equipment aligned and solo-run; Instruments and valves: Loop checks 3 instruments, 2 valves (range, alarms, trips, fail action) |
| Commissioning | Equipment and piping: In service on water (and VCM where it applies); Instruments and valves: Loops in AUTO / CAS at the design set points, tuned |
| Handover | Equipment and piping: Punch B closed, records signed; Instruments and valves: As-built loop records |

## SYS-03 Baffle and condenser cooling water

**Instrument list:** FT-120, TT-121, FT-130, TT-131, TT-120, FIC-120, FIC-130, VPC-130, FV-120, FV-130

| Stage | Checks |
|---|---|
| Mechanical completion | Instruments and valves: Installed and tagged: FT-120, TT-121, FT-130, TT-131, TT-120, FV-120, FV-130 |
| Pre-commissioning | Instruments and valves: Loop checks 5 instruments, 2 valves (range, alarms, trips, fail action) |
| Commissioning | Instruments and valves: Loops in AUTO / CAS at the design set points, tuned |
| Handover | Instruments and valves: As-built loop records |

## SYS-04 Reflux condenser, vents and recovery

**Equipment:** E-101, VP-101, C-101, D-103, D-106. **Instrument list:** TT-132, FI-135, FI-105, HIC-135, HIC-105, HV-135, PV-105, XV-105, XV-108, XV-109, XA-133, PSV-100

| Stage | Checks |
|---|---|
| Mechanical completion | Equipment and piping: Installed to the P&ID and the GA: E-101, VP-101, C-101, D-103, D-106; Equipment and piping: Supports, insulation, nameplates, punch A closed; Instruments and valves: Installed and tagged: TT-132, HV-135, PV-105, XV-105, XV-108, XV-109 |
| Pre-commissioning | Equipment and piping: Flushed / cleaned, leak tested; Equipment and piping: Rotating equipment aligned and solo-run; Instruments and valves: Loop checks 1 instruments, 5 valves (range, alarms, trips, fail action) |
| Commissioning | Equipment and piping: In service on water (and VCM where it applies); Instruments and valves: Loops in AUTO / CAS at the design set points, tuned |
| Handover | Equipment and piping: Punch B closed, records signed; Instruments and valves: As-built loop records |

## SYS-05 Charges: water + PVA, VCM, initiator, injection water

**Equipment:** TK-100, P-103, TK-107, D-104, P-104. **Instrument list:** FT-102, FT-101, FT-103, FQI-101, FQI-102, FIC-101, FIC-102, FIC-103, FV-101, FV-102, FV-103, XV-101, XV-102, XV-103

| Stage | Checks |
|---|---|
| Mechanical completion | Equipment and piping: Installed to the P&ID and the GA: TK-100, P-103, TK-107, D-104, P-104; Equipment and piping: Supports, insulation, nameplates, punch A closed; Instruments and valves: Installed and tagged: FT-102, FT-101, FT-103, FV-101, FV-102, FV-103, XV-101, XV-102, XV-103 |
| Pre-commissioning | Equipment and piping: Flushed / cleaned, leak tested; Equipment and piping: Rotating equipment aligned and solo-run; Instruments and valves: Loop checks 3 instruments, 6 valves (range, alarms, trips, fail action) |
| Commissioning | Equipment and piping: In service on water (and VCM where it applies); Instruments and valves: Loops in AUTO / CAS at the design set points, tuned |

| Stage | Checks |
|---|---|
| Handover | Equipment and piping: Punch B closed, records signed; Instruments and valves: As-built loop records |

## SYS-06 Short-stop and SIS

**Equipment:** D-105. **Instrument list:** TT-101, PT-101, XV-104, XS-101, HS-001

| Stage | Checks |
|---|---|
| Mechanical completion | Equipment and piping: Installed to the P&ID and the GA: D-105; Equipment and piping: Supports, insulation, nameplates, punch A closed; Instruments and valves: Installed and tagged: TT-101, PT-101, XS-101, HS-001, XV-104; SIS: SIS cabinet installed, cabling complete |
| Pre-commissioning | Equipment and piping: Flushed / cleaned, leak tested; Equipment and piping: Rotating equipment aligned and solo-run; Instruments and valves: Loop checks 4 instruments, 1 valves (range, alarms, trips, fail action); SIS: SIF-1 TSHH-101 / PSHH-101: end-to-end proof test; SIS: SIF-2 XS-101: end-to-end proof test; SIS: SIF-3 FSLL-110: end-to-end proof test; SIS: SIF-4 LSHH-100: end-to-end proof test |
| Commissioning | Equipment and piping: In service on water (and VCM where it applies); Instruments and valves: Loops in AUTO / CAS at the design set points, tuned; SIS: ESD test HS-001 on the water batch (CS-08) |
| Handover | Equipment and piping: Punch B closed, records signed; Instruments and valves: As-built loop records; SIS: SIS validation report (IEC 61511 cl. 15, typical), bypass register empty |

## SYS-07 Discharge and blowdown

**Equipment:** D-102, P-102. **Instrument list:** XV-106

| Stage | Checks |
|---|---|
| Mechanical completion | Equipment and piping: Installed to the P&ID and the GA: D-102, P-102; Equipment and piping: Supports, insulation, nameplates, punch A closed; Instruments and valves: Installed and tagged: XV-106 |
| Pre-commissioning | Equipment and piping: Flushed / cleaned, leak tested; Equipment and piping: Rotating equipment aligned and solo-run; Instruments and valves: Loop checks 0 instruments, 1 valves (range, alarms, trips, fail action) |
| Commissioning | Equipment and piping: In service on water (and VCM where it applies); Instruments and valves: Loops in AUTO / CAS at the design set points, tuned |
| Handover | Equipment and piping: Punch B closed, records signed; Instruments and valves: As-built loop records |

## SYS-08 DCS, I/O and alarms

| Stage | Checks |
|---|---|
| Mechanical completion | DCS: Cabinets, operator stations, network installed |
| Pre-commissioning | DCS: I/O check of 79 points; DCS: Alarm settings against the INL / ALM |
| Commissioning | DCS: Sequence (batch logic) run on water |
| Handover | DCS: Configuration backup, alarm rationalisation record (ISA-18.2) |

## SYS-09 Electrical and utilities (power, air, $N_2$, CW, steam)

**Instrument list:** PCV-141, M-101, P-101A, P-101B, P-103

| Stage | Checks |
|---|---|
| Mechanical completion | Motors: Motors connected: M-101, P-101A, P-101B, P-103; Utilities: Power, instrument air, $N_2$, CW, LP steam connected |
| Pre-commissioning | Motors: Insulation test, rotation, solo run; Utilities: Air receiver and dryer, $N_2$ pad PCV-141 set 1.5 bar g, CW and steam blown through |
| Commissioning | Motors: Coupled run under load; Utilities: Utilities on line for the water batch |
| Handover | Motors: Protection settings recorded; Utilities: Utility interfaces signed off |

# 5 Commissioning steps mapped to the OTS procedures and SOPs

| Step | Stage | What is done | OTS procedure | SOP | Where on the OTS |
|---|---|---|---|---|---|
| **CS-01** Mechanical completion walk-down and punch list | Mechanical completion | Walk every line against PVC-PID-001/002; punch items A (before pre-commissioning) and B; internal inspection of R-101 before box-up (polished wall, baffles, agitator clearance, thermowells). | — | SOP-R101-12 | P&ID tab (both sheets); SOP-R101-12 covers entry and box-up |
| **CS-02** Flushing and cleaning | Pre-commissioning | Water flushing of the charge, injection, jacket and CW lines through temporary strainers; air blowing of the $N_2$ and vent lines; clean the jacket loop until FT-110 strainers stay clean. | — | — | — |
| **CS-03** Leak test of R-101 with $N_2$ and vacuum hold test | Pre-commissioning | $N_2$ leak test at 9.0 bar g (the reference-batch peak 8.69 bar g rounded up; below PSHH-101 12.5 bar g and PSV-100 15 bar g) with every flange snooped; then evacuate to ≤ 0.05 bar a (the OTS cold-start target) and hold: pressure rise ≤ 5 mbar in 30 min (assumed). | Cold start: evacuate, charge, heat up and initiate (startup) | SOP-R101-02 | Exercises: cold start, step “R-101 evacuated below 0.05 bar a” |
| **CS-04** Instrument loop checks and I/O check | Pre-commissioning | Loop check every INL line from the device to the DCS faceplate (66 tags): range and units against the instrument list, alarm and trip settings, fail action of each valve with the air off. | — | SOP-R101-01 | Operate tab faceplates; Instrumentation & control tab (loop explorer) |
| **CS-05** SIS site acceptance and proof test of every SIF | Pre-commissioning | Proof test each SIF end to end (sensor, logic, final element) before VCM is on site; test the reset and the restart after a trip. Interval T = 1 year (PFDavg with T = 1 yr). Full stroke of XV-104 with the pot isolated, D-105 pressure and charge check, TE-101 / PT-101 calibration, motor-run and FT-110 trip tests, LSHH-100 level test. | — | SOP-R101-01, SOP-R101-09 | Batch & SIS tab: trip, first-out, reset; SRS of SIF tab |
| **CS-06** Motor solo runs and pump runs on water | Pre-commissioning | Solo run (uncoupled) and rotation check of M-101, P-101A, P-101B, P-103; coupled runs of P-101A/B on the water-filled jacket loop; planned switch-over P-101A → P-101B on the running loop. | Jacket pump fails: keep the batch (esd) | SOP-R101-10 | Exercises: the jacket-pump procedure; SOP tab SOP-R101-10 |
| **CS-07** Water batch | Commissioning | Charge 60 t of water through FIC-101 / FQI-101, start M-101, heat on steam through TV-110A to 57 °C, put TIC-100 → TIC-110 in cascade and check the split range; step the set point and tune; run FIC-120, FIC-130 and LIC-100 / FIC-103; discharge through XV-106. The same recipe without VCM tests every utility and loop at operating temperature. | Cold start: evacuate, charge, heat up and initiate (startup) | SOP-R101-03, SOP-R101-05, SOP-R101-07 | Exercises: cold start (water, agitator, heat-up and cascade steps); Control loop design tab (tuning) |
| **CS-08** ESD and short-stop test on the water batch | Commissioning | With water at 57 °C and the short-stop pot D-105 filled with water: push HS-001 and check the C&E effects (short-stop XV-104 injects, charges close, TV-110B / FV-120 / FV-130 forced open, XV-108 to D-103), the first-out and the reset. Operators practise the manual short-stop. | Initiator slug: stop a runaway before the SIS does (runaway) | SOP-R101-08 | Exercises: stop a runaway before the SIS does; Batch & SIS tab |
| **CS-09** Inerting and introduction of VCM | Commissioning | Evacuate R-101 to ≤ 0.05 bar a and close XV-109; the VCM permissive needs PI-100 below the water vapour pressure + 0.15 bar (0.32 bar a at 55 °C): with air as the residual gas that is at most 32 mbar of $O_2$. First VCM to the charge line under a vapour leak test, then the first VCM charge through FIC-102 / FQI-102 (51 t). | Cold start: evacuate, charge, heat up and initiate (startup) | SOP-R101-02, SOP-R101-04 | Exercises: cold start (evacuation and VCM charge steps); SOP tab SOP-R101-02 and -04 |
| **CS-10** First batch | Commissioning | Heat-up, initiator, reaction with E-101 brought into service and inerts vented, end point at PD-100 1.8 bar, short-stop, degassing below 2.0 bar g, $N_2$ pad and discharge. The cold batch on the Transition tab takes 7.75 h from empty to empty (initiation at 1.37 h, end point at 6.64 h). | Cold start: evacuate, charge, heat up and initiate (startup); End of batch: end point, short-stop, degas and discharge (end); Reflux condenser performance falls (unknown upset) (inerts) | SOP-R101-05, SOP-R101-06, SOP-R101-07, SOP-R101-11 | Exercises: cold start, end of batch, condenser performance; SOP tab |
| **CS-11** Performance test | Handover | Three consecutive batches at the reference recipe (assumed); the guarantees in section 4 measured from the DCS and the laboratory; cooling-water optimisation once the guarantees are met. | Cooling-water optimisation (live, 45 min) (opt) | SOP-R101-06 | Exercises: cooling-water optimisation; Efficiency tab |
| **Step** | **Stage** | **What is done** | **OTS procedure** | **SOP** | **Where on the OTS** |
| **CS-12** Handover to operations | Handover | Punch list B closed, SOP register approved, operators trained and graded on every OTS procedure, as-built drawings and documents (INL, C&E, SRS) at the final revision. | Cold start: evacuate, charge, heat up and initiate (startup); End of batch: end point, short-stop, degas and discharge (end); Jacket pump fails: keep the batch | — | Exercises (every procedure graded), Workbook |

| Step | Stage | What is done | OTS procedure | SOP | Where on the OTS |
|---|---|---|---|---|---|
| | | | (esd); Initiator slug: stop a runaway before the SIS does (runaway); Reflux condenser performance falls (unknown upset) (inerts); Cooling-water optimisation (live, 45 min) (opt) | | |

## 5.1 SIF proof tests (CS-05, before VCM)

| SIF | Trip | Sensor | Setting | Test |
|---|---|---|---|---|
| **SIF-1** | TSHH-101 / PSHH-101 | TE-101 and PT-101, own elements (not TE-100 / PT-100); either one trips (1oo2 diverse) | > 63 °C (10 s) or > 12.5 bar g (5 s) | Force TSHH-101 / PSHH-101 (> 63 °C (10 s) or > 12.5 bar g (5 s)) at the sensor (TE-101 and PT-101); check the trip and first-out on the SIS console and that XV-104, XV-101, XV-102, XV-109, FIC-103, TV-110A, TV-110B, FV-120, FV-130 move; reset only when the cause has cleared |
| **SIF-2** | XS-101 | M-101 motor-run contact, 1oo1 (does not see a broken coupling) | Motor stopped while reacting | Force XS-101 (Motor stopped while reacting) at the sensor (M-101 motor-run contact); check the trip and first-out on the SIS console and that XV-104, XV-101, XV-102, XV-109, FIC-103, TV-110A, TV-110B, FV-120, FV-130 move; reset only when the cause has cleared |
| **SIF-3** | FSLL-110 | FT-110, 1oo1 (shared with the FI-110 alarm and the batch permissive) | < 120 m³/h | Force FSLL-110 (< 120 m³/h) at the sensor (FT-110); check the trip and first-out on the SIS console and that XV-104, XV-101, XV-102, XV-109, FIC-103, TV-110A, TV-110B, FV-120, FV-130 move; reset only when the cause has cleared |
| **SIF-4** | LSHH-100 | Own level measurement in the simulator, 1oo1 (drawn on LT-100 on PVC-PID-001) | > 94 % | Force LSHH-100 (> 94 %) at the sensor (Own level measurement in the simulator); check the trip and first-out on the SIS console and that XV-101, XV-102, FIC-103 move; reset only when the cause has cleared |

## 5.2 Worked calculation: oxygen allowed by the VCM permissive

Evacuated to 0.05 bar a at 30 °C, the 141.5 m³ reactor holds 281 mol of gas, 0.059 kmol of it $O_2$. The model inhibits the rate by $f = 1/(1 + (n/0.002)^2) = 0.0011$ at initiation, so the batch waits about 12 min while the radicals consume the $O_2$. The permissive (0.15 bar of inerts in the 80.8 m³ gas space at 55 °C) allows up to 0.093 kmol $O_2$ if the inerts are air.

## 6 Performance test

Three consecutive batches at the reference recipe (assumed). Guarantee values are the model's reference batch (pvBatch from the start of the reaction to the PD-100 end point).

| Item | Reference batch | Unit | Acceptance | Measured by | DCS calculation |
|---|---|---|---|---|---|
| **Reaction time, initiation to end point** | 5.27 | h | ≤ +5 % (assumed) | XV-103 open → PD-100 ≥ 1.8 bar | PD-100 (running maximum of PI-100 minus PI-100) |
| **Conversion at the end point** | 0.848 | – | ≥ 0.84 (assumed) | XI-100; FQI-102, dried PVC weight (lab) | XI-100 calorimetric conversion |
| **PVC per batch** | 43.24 | t | ≥ 42.5 t (assumed) | FQI-102 × XI-100; dryer weigh scale | XI-100 |
| **K-value** | 66.99 | – | 66–68 (grade band) | Laboratory viscosity (K-value); KX-100 | KX-100 estimate |
| **Temperature control after 15 min** | 0.33 | °C (max deviation) | ≤ 0.5 °C | TT-100, TT-101 | — |
| **Peak heat release (at X ≈ 0.72)** | 3.43 | MW | removed with TV-110B ≤ 80 % | QI-100, TV-110B position | QI-100 = Σ flow × cp × ΔT (FT-110/TT-110/TT-111, FT-120/TT-120/TT-121, FT-130/TT-131) |
| **Peak heat removed** | 3.54 | MW | ≥ peak release | QI-100 | QI-100 |
| **Jacket CW valve TV-110B, maximum opening** | 54.70 | % | ≤ 80 % (trim margin) | TIC-110 output | TY-110 split range |
| **Peak pressure** | 8.69 | bar g | below PI-100 H alarm 11 bar g | PT-100 | — |
| **Agitator peak power** | 338 | kW | ≤ motor rating 330 kW | JT-100 (JI-100) | — |
| **Cooling water per batch** | 1615 | m³ | record (no guarantee) | FT-120 + FT-130 totals; jacket injection estimated | Totals in the DCS (to be configured) |
| **Agitator energy per batch** | 1579 | kWh | record | JT-100 | Integral of JI-100 |
| **Coarse fraction (lumps)** | 0.00 | % | ≤ 0.5 % (assumed) | Laboratory sieve | — |

## 7 Checks against the model and the other documents

9 of 10 checks pass.

| # | Check | Against | Result | Detail |
|---|---|---|---|---|
| 1 | Every OTS procedure (PROCS) mapped to a commissioning step | PROCS keys | ✓ | startup → CS-03/CS-07/CS-09/CS-10; end → CS-10; esd → CS-06; runaway → CS-08; inerts → CS-10; opt → CS-11 |
| 2 | Every SOP of the register mapped to a commissioning step | SOP tab (spMake) | ✓ | 12 SOPs |
| 3 | Every equipment item of PFD-100 in a procurement package | PFD-100 (PF_SPEC.equipment) | ✓ | 18 items in 6 packages |
| 4 | Every SIF has a proof test in pre-commissioning | PVC-V1-IN-SRS-001 (SF.srs) | ✓ | SIF-1 TSHH-101 / PSHH-101; SIF-2 XS-101; SIF-3 FSLL-110; SIF-4 LSHH-100 |
| 5 | Every INL tag allocated to a completion system | PVC-V1-IN-INL-001 (EI_INL) | ✓ | 66 tags in 9 systems |
| 6 | $N_2$ leak-test pressure above the batch peak and below the trip and the PSV | K.Phh, K.Ppsv, reference batch | ✓ | 9 bar g vs peak 8.69, PSHH-101 12.5, PSV-100 15 bar g |
| 7 | Evacuation target inside the VCM permissive | purgePermissives(), PROCS startup | ✓ | target 0.05 bar a (OTS cold start) vs permissive 0.32 bar a (water vapour at 55 °C + 0.15 bar) |
| 8 | Performance guarantees inside the alarm limits | ALM (PI-100 H, TIC-100 L/H) · reference batch | ✓ | peak 8.69 bar g < 11; 56.72–57.33 °C inside 56–58 °C |
| 9 | Agitator power in the reference batch within the motor rating | PVC-PID-001 (M-101 330 kW) · IND JI-100 · reference batch | ✗ | peak absorbed 338 kW vs motor 330 kW (design ≈290 kW at 2 h) |
| 10 | DCS and SIS packages sized from the I/O count | PVC-V1-IN-INL-001 I/O count | ✓ | 79 points before spare; 97 installed with 20 % spare |

## 8 Engineering queries

| No. | Subject | Finding | Raised against | Proposed |
|---|---|---|---|---|
| EC-01 | **Agitator motor rating** | In the reference batch the agitator absorbs up to 338 kW near the end of the batch, above the 330 kW motor drawn on PVC-PID-001 (≈290 kW at | PVC-PID-001, M-101 datasheet, simulator model (agitator power) | Confirm the absorbed-power curve with the agitator vendor; size the motor on the end-of-batch viscosity |

| No. | Subject | Finding | Raised against | Proposed |
|---|---|---|---|---|
| | | the 2 h design point). | | (e.g. 375 kW) or accept a short overload in the motor datasheet; agree the performance-test limit. |
| EC-02 | **Cooling water to the jacket not metered** | The CW injected into the jacket loop through TV-110B (and spilled through PCV-112) has no flowmeter; the performance test can only estimate it from QI-100 and the baffle and condenser meters. | PVC-PID-002, PVC-V1-IN-INL-001 | Add a flow element on the CW injection or the spill line if the CW consumption becomes a guarantee; otherwise record it as an estimate. |
| EC-03 | **Procedure key “esd”** | The OTS procedure with key “esd” is the jacket-pump switch-over (keep the batch), not an emergency shutdown; the commissioning plan maps it to the pump runs. | OTS procedure list (PROCS) | Rename the key (e.g. “pumpfail”) at the next OTS revision; keep the title. |
| EC-04 | **Oxygen acceptance by pressure only** | R-101 has no oxygen analyser: the acceptance before VCM is the evacuation pressure (permissive 0.32 bar a). Commissioning has no independent check of the residual $O_2$ after the first evacuation. | PVC-V1-IN-INL-001, SOP-R101-02 | Use a portable $O_2$ analyser on the vacuum line for the first batches (commissioning only), or add an $O_2$ analyser on the vacuum-pump discharge. |